FIPS 140-3 (Cryptographic Module Validation)

Anticloud FZ LLE — PAX L5 Narrow L2 General 27B

Run: 2026-10-02T09:42:57.025910Z | Framework ID: 14_FIPS_140_3

Standard

FIPS 140-3 (NIST, 2019), ISO/IEC 19790:2012

Result

PARTIAL — SHA3-256 (FIPS-approved), AES-256-GCM (FIPS-approved); scrypt key derivation not FIPS-approved

Score

Level 2 target — SHA3/AES approved; scrypt requires substitution with PBKDF2/HKDF for full FIPS

Notes

SHA3-256 (FIPS 202), AES-256-GCM (SP 800-38D) are FIPS-approved. Recommendation: replace scrypt with PBKDF2-SHA512.

Real Benchmark Numbers (Kaggle T4, loiskleinner/pax-benchmark-runner v3)

Module	Key Metric	Value
CRDT	lww_set_us	0.51
CRDT	merge_10k_ms	0.07
CRDT	convergence	GUARANTEED
CryptographicProvenance	write_per_receipt_ms	0.059
CryptographicProvenance	verify_1k_chain_ms	2.53
ZeroTrustLocalMesh	ca_gen_ms	367.5
ZeroTrustLocalMesh	cert_issue_ms	333.8
ZeroTrustLocalMesh	tls_version	1.3
LocalObservabilityStack	spans_per_sec	17558
LocalObservabilityStack	logs_per_sec	24492
SovereignMemory	write_per_entry_ms	51.94
SovereignMemory	encryption	AES-256-GCM
AIOSS_AuditChain	entries_per_sec	232665

| AIOSS_AuditChain | hash_algo | SHA3-256 |

Environment Provenance (Preprioception)

Field	Value
Run ID	loiskleinner/pax-benchmark-runner v3
Run Date	2026-10-02
Platform	Kaggle T4 x2 GPU, Ubuntu 22.04, Python 3.12
GPU	NVIDIA Tesla T4 (2× 16 GB VRAM)
CPU	Intel Xeon (2 vCPU, 13 GB RAM)
Inference Temperature	N/A (code benchmarks, no LLM sampling)
KV Cache	N/A (no LLM inference in this cell)
Framework	llama-cpp-python 0.3.x (installed from source with CUDA)

Cryptographic Integrity

Field	Value
SHA3-256 (content)	16055c2d7559282dd803f09a0894518dcc392004198fc069d25930d28cda702d
SHA3-256 (chain)	8afd0dbc03c0363f0a0d664039f0ef8bbe9ebb6e101e6b97e48845297062764e
Timestamp UTC	2026-10-02T09:42:57.025910Z
Hash Algorithm	SHA3-256 (FIPS 202 compliant)

Anticloud FZ LLE · PAX L5 Narrow L2 General 27B · AIOSS Ledger Genesis: 8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560